



RESEARCH PAPER

SmoIVLM

Redefining Small and Efficient Multimodal Models



Compact Vision-Language Models for Edge Deployment

256M / 500M / 2.2B Parameters

Sub-1GB RAM Inference

Large VLMs Are Powerful but Impractical for Edge Deployment

The Challenge

State-of-the-art VLMs require massive compute and memory, making them impractical for mobile and edge device deployment.

Memory Footprint Comparison

EXISTING MODELS

Molmo-A1B-7B

27.7 GB

GPU RAM Required

IMPOSSIBLE ON EDGE

Idefics-80B

80B

Parameters (300x larger)



SMOLVLM SOLUTION

SmoIVLM Family

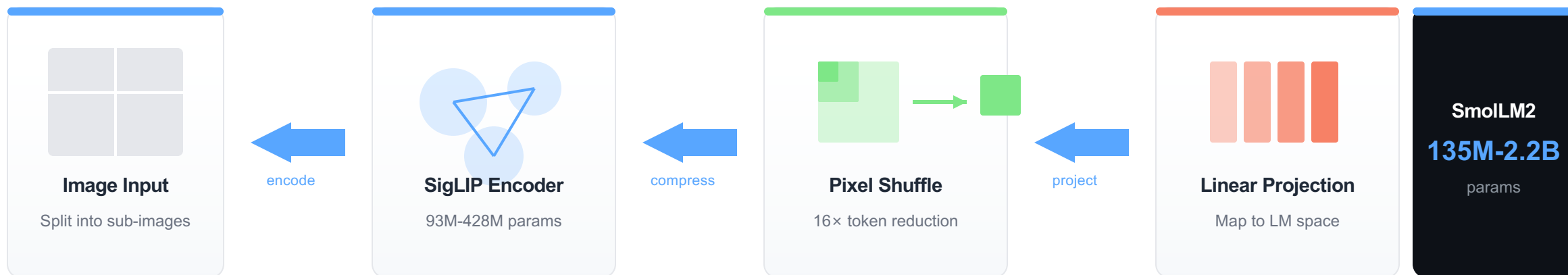
0.8-4.9 GB

Sub-1GB to 4.9GB RAM

Key Insight: SmoIVLM-256M runs inference with less than 1GB RAM and outperforms the 300× larger Idefics-80B

SmoIVLM Architecture

Image Splitting → Vision Encoder → Pixel Shuffle → SmoLM2



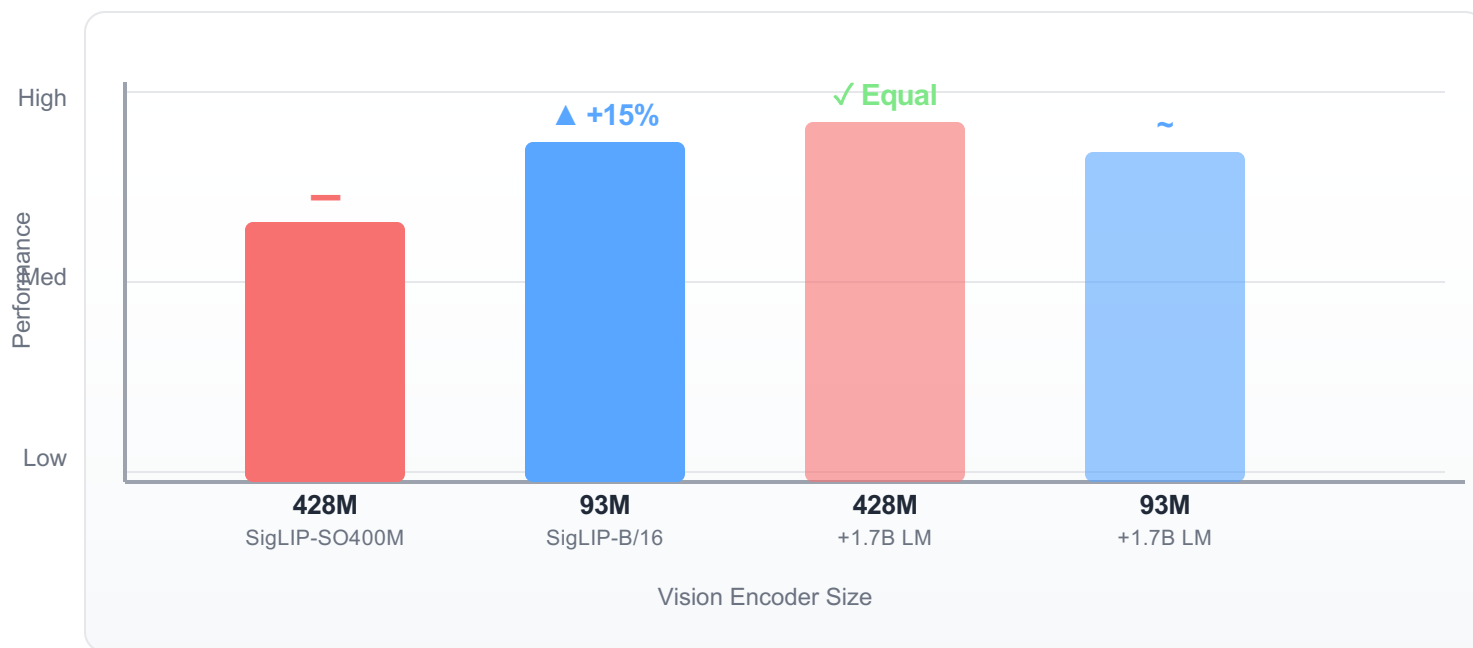
Three Key Architectural Innovations

- 1 Balanced Encoder-LM Compute**
Smaller encoders (93M) outperform larger ones (428M) for compact VLMs
- 2 Extended Context Length**
2K → 16K tokens with RoPE base adjustment (10K → 273K)
- 3 Aggressive Token Compression**
Pixel shuffle (r=4) gives 16× token reduction for small VLMs

Smaller Vision Encoders Outperform Larger Ones in Compact VLMs

Insight: Balanced encoder-LM compute allocation is critical for small VLMs. The 93M SigLIP-B/16 paired with 135M LM outperforms the 428M SigLIP-SO400M with the same LM.

Encoder Performance Comparison (135M LM)



CONCLUSION

For smallest language models:

93M encoder > 428M encoder

- Larger encoder only justified at 1.7B+ LM scale

PARAMETER OVERHEAD

10%

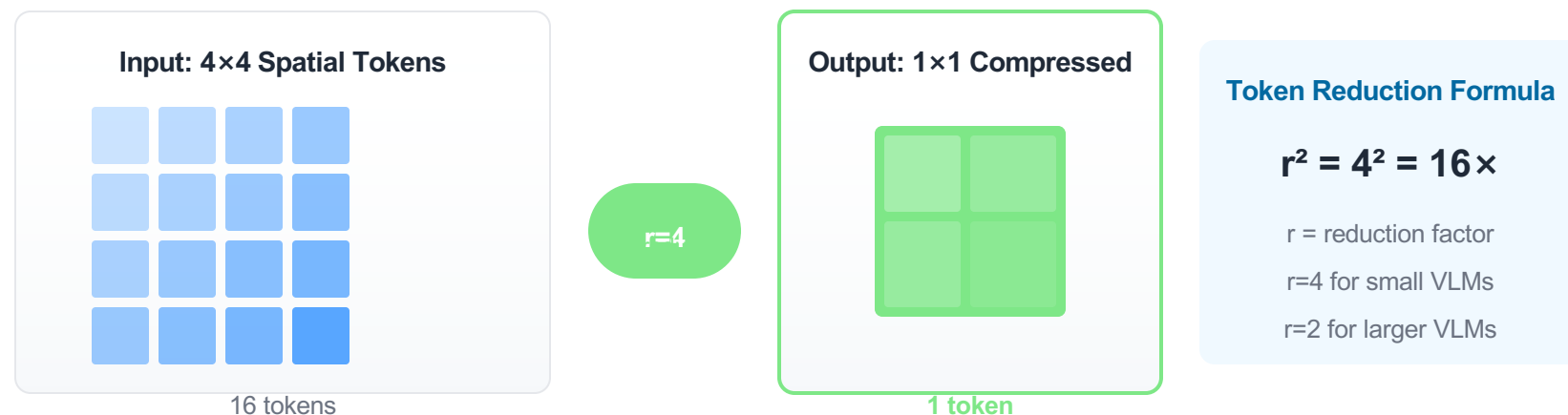
encoder param overhead for
SigLIP-SO400M vs SigLIP-B/16

KEY FINDING 3

Aggressive Pixel Shuffle (r=4)

Cuts Visual Tokens 16× for Small Models

Pixel Shuffle Mechanism (Space-to-Depth)



Benefits for Small VLMs



Reduced Attention Overhead

Fewer tokens = less compute in self-attention layers



Better Long-Context Modeling

More context room for higher resolution images and video



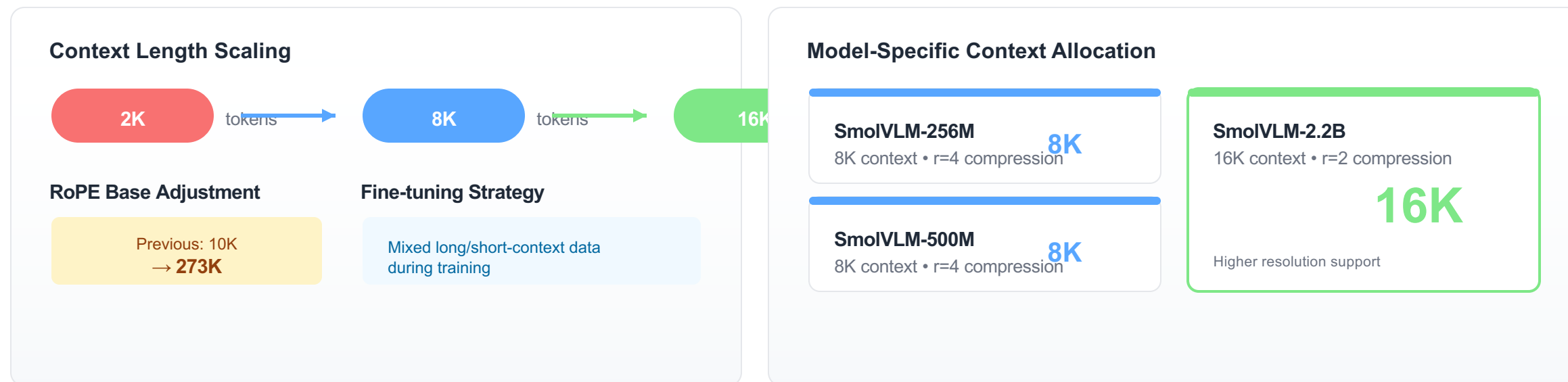
Memory Efficiency

Enables sub-1GB inference on resource-constrained devices

Trade-off: Larger VLMs (2.2B) use $r=2$ because they have more compute budget and benefit from finer-grained visual representation, while smaller VLMs prioritize efficiency over granularity.

Extending Context from 2K to 16K Tokens

Unlocks Higher Image Resolution



Why Extended Context Matters

Higher Image Resolution

More tokens = process larger images without downsampling

Video Understanding

Longer context enables processing more video frames simultaneously

Multi-Image Reasoning

Support for multiple images in a single context window

Competitive Performance at Sub-1GB to 4.9GB RAM



Benchmark Results (13 datasets)

Category	Benchmark	256M	500M	2.2B	Best OS
Single-Image	OCRBench	52.6%	61.0%	72.9%	54.7% (Molmo)
Single-Image	AI2D	46.4%	59.2%	70.0%	71.0% (Molmo)
Single-Image	ChartQA	55.6%	62.8%	68.7%	48.0% (Molmo)
Single-Image	TextVQA	50.2%	60.2%	73.0%	61.5% (Molmo)
Single-Image	DocVQA	58.3%	70.5%	80.0%	77.7% (Molmo)
Single-Image	ScienceQA	73.8%	80.0%	89.6%	87.5% (Molmo)
Multi-task	MMMU	29.0%	33.7%	42.0%	33.9% (Molmo)
Multi-task	MathVista	35.9%	40.1%	51.5%	37.6% (Molmo)
Multi-task	MMStar	34.6%	38.3%	46.0%	43.1% (Molmo)

Key Result: SmoIVLM-256M (0.8GB) surpasses the 300× larger Idefics-80B, demonstrating architectural efficiency outperforms raw scale.

Average	All Benchmarks	44.0%	51.0%	59.8%	—
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Real-Time Inference on iPhones

Up to 80 Tokens/Second on Apple Silicon

HuggingSnap iOS App

- Fully on-device processing — no cloud dependency
- Processes photos and videos entirely locally

Max Throughput (256M)

~80 tokens/s

Batch=64 (High-end)

~1000
tokens/s

Throughput by Device

MacBook Pro M1 Max 64GB

High-end desktop

~80 tok/s

MacBook Air M3 16GB

Consumer laptop

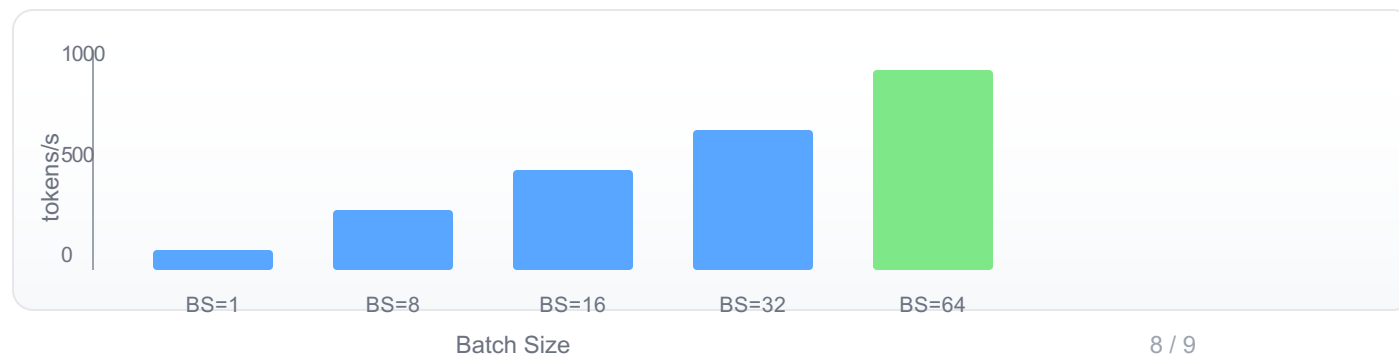
~55 tok/s

iPhone 15 Pro (On-Device)

Mobile device

~30 tok/s

Linear Scalability with Batch Size



KEY OBSERVATIONS

- 1. Linear scaling:** Throughput scales roughly linearly with batch size for smallest model
- 2. Mobile feasible:** ~30 tokens/s on iPhone 15 Pro
- 3. Zero cloud:** Complete privacy — no data leaves device

CONCLUSION

Key Takeaways

Design Principles for Efficient Multimodal Models

1

Balance Encoder-LM Compute

Smaller vision encoders (93M) outperform larger ones (428M) for compact VLMs — allocate params wisely

2

Extend Context Length

2K → 16K tokens with RoPE base adjustment (10K → 273K) enables higher resolution and video understanding

3

Use Aggressive Token Compression

Pixel shuffle ($r=4$) gives $16\times$ token reduction for small models eases attention overhead and improves efficiency

4

Sub-1GB Inference is Achievable

SmoVLM-256M runs at 0.8GB RAM — deployable on mobile and edge devices without cloud dependency

5

Fully Open-Source

All models, datasets, code, and HuggingSnap mobile app available for reproducibility and research

IMPLICATION

Architectural efficiency matters more than raw scale. SmoVLM-256M surpasses $300\times$ larger Llafacs-80B.

THANK YOU

Questions & Discussion

arXiv:2504.05299 | Hugging Face | Stanford University